

Active Inference World Models: From Embodied Control to General-Purpose Adaptive Intelligence

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Abstract

World models have become a central concept in artificial intelligence, robotics and cognitive science because adaptive agents must infer hidden causes, predict future states and use internal models to guide action under uncertainty. However, many contemporary world-model approaches treat perception, planning, control, uncertainty and goal-directed behaviour as partially separable components. This paper argues that active inference provides a principled framework for building world models that are intrinsically embodied, action-oriented and uncertainty-sensitive. Under active inference, an agent does not first learn a model of the world and then use it for control. Instead perception, action, policy selection and learning are coupled through the minimisation of variational and expected free energy. Here we develop the concept of active inference world models as generative architectures that integrate sensory inference, latent dynamics, semantic priors, epistemic exploration and preference-guided action. Building on recent work on embodied active inference control, we show how a ray-cast PyBullet agent can be extended from reactive sensorimotor coordination toward a structured belief system over rooms, objects, candidate viewpoints and hidden object locations. In this setting, the agent uses semantic priors to guide search, update its beliefs through positive and negative evidence and select actions that balance pragmatic goal fulfilment against epistemic uncertainty reduction. We argue that this provides a concrete route from embodied control toward more general forms of adaptive intelligence, in which agents do not merely predict the world, but actively interrogate, model and transform it through action.

1 Introduction to World Models

As artificial intelligence systems move from narrow pattern recognition toward agents that must act, plan and adapt in changing environments, the question of how such systems should represent

the world has become increasingly central. A classifier can map inputs to labels, a language model can predict the next token and a reactive controller can transform sensory signals into actions, but an adaptive agent requires to infer the hidden causes of sensory input, predict how the world might change, estimate the consequences of its own actions and revise its beliefs when observations violate expectation. In this broad sense, a world model is not simply a compressed representation of sensory data, but an internal generative structure that allows an agent to use the past and present to anticipate possible futures.

This idea has a long history in cognitive science, control theory and neuroscience, although it has recently become prominent in artificial intelligence through model-based reinforcement learning, latent dynamics models and architectures for autonomous agents [Ha and Schmidhuber, 2018, Hafner et al., 2023, LeCun, 2022]. In contemporary AI, world models are attractive because they offer a route beyond purely reactive behaviour, thus, an agent can learn a model of environmental dynamics and use that model to simulate, evaluate and improve behaviour internally. This shift is especially important in settings where interaction is costly, sparse rewards are difficult to obtain or long-horizon consequences are not immediately observable because the agent must be able to reason about what would happen under counterfactual actions.

The recent success of world-model approaches in reinforcement learning has shown that learned latent dynamics can support impressive generalisation across control tasks, particularly when agents are trained to predict future observations, rewards and continuation signals from compact internal states [Hafner et al., 2023]. However, these approaches also reveal a deeper conceptual issue. In many current systems, the world model is treated as one component within a broader engineering pipeline. In this, perception is handled by an encoder, dynamics by a latent transition model, value by a reward predictor, planning by a policy optimiser and uncertainty by auxiliary heuristics or regularisation terms. This modularity is practical and often powerful. But because these components share no common currency, the system has no internal mechanism for recognising when its own representations are unreliable.

This paper argues that active inference provides a principled way to couple prediction, uncertainty, preference and action within a single generative architecture. Under active inference, perception, action, learning and policy selection are coupled through the minimisation of variational free energy and expected free energy [Friston et al., 2016, Parr and Friston, 2019, Friston et al., 2017]. The agent acts to fulfil predictions, but those predictions are constrained by a generative model that links hidden states, observations, actions and prior preferences. Behaviour therefore emerges from an attempt to reduce uncertainty about the causes of sensation while also realising preferred outcomes, meaning that epistemic exploration and goal-directed control are complementary terms within the same objective function.

This distinction matters because an embodied world model must do more than forecast sensory trajectories. It must decide which observations are worth sampling, which parts of the environment are currently uncertain, which actions would be informative, which futures are preferred and how much confidence should be assigned to each belief or policy. A robot searching for an object in a house, for example, must infer that rooms contain objects, that objects tend to occur in semantically meaningful locations, that failing to see an object from one viewpoint provides negative evidence and that moving to a new viewpoint may be useful because it resolves uncertainty. These are

central features of intelligent behaviour.

We use the term *active inference world model* to describe a generative architecture in which these components are integrated explicitly. Such a model contains beliefs about hidden states of the world, beliefs about the consequences of action, prior preferences over outcomes, precision estimates that determine confidence in sensory evidence and policies and epistemic terms that make information-seeking behaviour a natural part of action selection. This framework treats the world model as an embodied belief system that is continuously updated through action. The agent does not only predict what will happen next but it actively samples the world to test, refine and exploit its own generative model.

The aim of this paper is to develop this idea conceptually and computationally. We first review the gaps in current world-model AI, focusing on the separation between prediction, control, uncertainty and value. We then introduce active inference as a theory of world modelling, emphasising the role of generative models, variational free energy, expected free energy, precision and prior preferences. Building on recent work on embodied active inference control, we then describe how a simple ray-cast PyBullet agent can be extended from reactive sensorimotor coordination toward a more detailed belief system over rooms, objects, candidate viewpoints and hidden object locations. Thus, active inference provides a route from embodied control to general-purpose adaptive intelligence because it treats world modelling, exploration and action as mutually dependent aspects of the same inferential architecture.

2 Limitations of current world-model AI

Although recent world-model systems have made substantial progress [Ha and Schmidhuber, 2018], especially in model-based reinforcement learning, a set of conceptual and practical problems remains unresolved. Most current approaches position the world model as a predictive engine within a broader control architecture. Perception, uncertainty, action and preference are then defined separately and coordinated around it. This becomes increasingly consequential as agents are expected to operate outside closed benchmark environments, where goals may be underspecified, observations may be ambiguous and the value of an action may depend on whether it reduces uncertainty or immediately increases reward.

A first limitation concerns the relationship between prediction and action. In many world-model approaches, the model is trained to predict future sensory states or latent features, while the policy is trained separately to maximise expected reward under those predictions [Ha and Schmidhuber, 2018, Hafner et al., 2023]. This separation is computationally useful, but it can obscure the fact that action is not solely a downstream of prediction. In embodied agents, action changes the data that the agent receives, meaning that the agent actively creates the sensory conditions under which inference takes place. A model that predicts future observations without also representing the epistemic consequences of action risks treating perception as an input to control.

A second limitation concerns uncertainty. World-model agents often represent uncertainty implicitly through stochastic latent states, ensembles, distributional predictions or exploration bonuses, but uncertainty is not always treated as a first-class quantity that determines both belief updating

and policy selection. Some uncertainty reflects ignorance that can be reduced by sampling the world, some reflects volatility that should change the rate of learning and some reflects noise that should be discounted. An agent searching a room for an object, for example, should distinguish between not knowing whether the object is present, not knowing whether its sensor is reliable and knowing that a particular viewpoint has already provided useful negative evidence. Without an explicit account of these distinctions, exploration can become either too random, too reward-driven or too dependent on task-specific heuristics.

A third limitation is that reward-based formulations can make goal-directed behaviour appear more externally specified than it is in natural agents. Reinforcement learning typically defines behaviour in relation to scalar rewards, which are powerful for training but can be difficult to specify in open-ended environments because the reward function must stand in for preferences, constraints, information value and task context [Amodei et al., 2016]. In practice, this often means that the agent is optimised for a proxy signal when it should be for an explicitly represented set of preferred states. This is not simply a technical inconvenience because misspecified rewards can produce brittle or unintended behaviour, while sparse rewards can leave the agent with little guidance about which uncertainties to resolve before a reward becomes available [Amodei et al., 2016, Hadfield-Menell et al., 2016]. A world model intended for adaptive intelligence therefore needs a richer account of why some futures are preferable to others and how those preferences interact with uncertainty reduction during action selection.

A fourth limitation concerns semantic structure. Contemporary latent world models can learn powerful compressed representations, but the resulting latent states are not always organised around the kinds of entities, relations and affordances that support flexible behaviour. For an embodied agent, the relevant structure of the world is not exhausted by visual similarity or short-term predictability. Rooms, objects, routes, viewpoints and tasks have functional meaning because they constrain what can be done, what can be inferred and what should be expected. A mug, for example (see section 4), is an object that is more likely to be found in some places than others, can be occluded, can be searched for and can change the value of future actions. If world models are to support general-purpose adaptation, they need mechanisms for linking low-level sensory evidence to higher-level beliefs about hidden causes, task structure and context-dependent affordances.

A fifth limitation is that many current systems remain weakly grounded in the active nature of perception. In benchmark settings, the environment is often treated as a source of observations that are delivered to the agent at each timestep, while the policy determines how the agent moves through a pre-defined state space. However, in embodied settings, perception depends on where the agent looks, how it moves, what it expects to find and what it has already ruled out. Negative evidence is especially important here because failing to observe an object from a particular location should change the agent’s beliefs about where that object is likely to be. This form of inference is difficult to capture if the world model is mainly a predictor of future frames, but it becomes natural if the agent maintains explicit beliefs about hidden states and updates those beliefs through action-dependent observations.

These limitations suggest that the next step for world-model AI is to develop architectures in which prediction, uncertainty, preference and action are formally coupled. Such architectures should be able to represent hidden causes, select actions because they are informative as well as instrumen-

tally useful, update beliefs using both positive and negative evidence and modulate confidence in perception and policy depending on context. In this sense, the missing ingredient is a generative account of embodied agency in which the world model is the structure through which the agent perceives, acts and learns.

3 Active inference as a theory of world modelling

Active inference begins from the assumption that an adaptive system must maintain itself within a limited range of viable states, despite being coupled to an environment that is only partially observed and continuously changing [Friston, 2010, Friston et al., 2016]. This framing immediately makes world modelling central because the agent cannot act directly on the hidden causes of its sensory input. It must infer those causes through a generative model that specifies how latent states of the world give rise to observations, how those states evolve over time and how actions change the sensory evidence that becomes available. In this sense, active inference is not a theory in which world modelling is added to a controller, but a theory in which perception and action are both consequences of maintaining and using a generative model.

The central quantity in this framework is variational free energy, which measures how well the agent’s current beliefs explain its sensory observations under its generative model [Friston, 2010, Bogacz, 2017]. More formally, free energy provides a tractable upper bound on surprise, meaning that the agent can reduce a computable proxy for how unexpected its observations are, even when the true probability of those observations cannot be evaluated directly.

Perception corresponds to the updating of approximate posterior beliefs about hidden states, parameters or causes, while learning corresponds to slower changes in the model itself. This gives active inference an explicitly inferential account of perception because it explains sensory input by inferring the hidden states that would make those observations likely. A world model, in this setting, is therefore a structured hypothesis about the causes of sensation rather than a passive store of predictive features.

Action enters the framework because the agent can also reduce expected uncertainty by sampling the world in ways that make its observations conform to its predictions. This does not mean that the agent simply imposes arbitrary predictions on the environment because those predictions are constrained by prior preferences, learned dynamics and sensory evidence. Instead action is selected to bring about observations that are both expected under a policy and consistent with preferred outcomes. The same generative model that supports perceptual inference thus also supports control because the agent evaluates possible policies in terms of their expected consequences for future beliefs and future observations [Friston et al., 2017, Parr et al., 2022].

Expected free energy extends this logic from perception to planning. Whereas variational free energy concerns the explanation of current sensory data, expected free energy scores policies according to the beliefs and observations they are expected to produce in the future [Friston et al., 2015, Friston et al., 2017]. This quantity is often decomposed into terms that correspond to pragmatic value (policies are preferred when they are expected to realise desired outcomes) and epistemic value (policies are also preferred when they are expected to reduce uncertainty about hidden states

or model parameters). The important point for world models is that exploration follows from the value of resolving uncertainty under the generative model.

This makes active inference especially relevant for embodied world modelling because the agent’s model must contain the conditions under which observations become available. A viewpoint, movement or search action is not only a step through space, but an intervention on future sensory evidence.

The framework also provides a natural role for precision, which determines the confidence assigned to prediction errors, sensory modalities, beliefs and policies [Friston et al., 2012, Parr et al., 2022]. Precision matters because an agent must decide how strongly it should rely on different sources of evidence. A noisy sensor should not dominate belief updating, an uncertain policy should not strongly control behaviour and a highly reliable observation should be able to rapidly reshape the posterior distribution over hidden states. In computational terms, precision modulation determines the gain on inference and action selection, while in functional terms, it allows the agent to adapt its confidence to context rather than treating all observations and predictions as equally informative.

This perspective reframes the meaning of a world model. In model-based reinforcement learning, the world model is often understood as a learned simulator that predicts future observations, latent states or rewards. In active inference, the world model is broader because it includes an observation model, a transition model, prior preferences, beliefs about policies and precision-weighted uncertainty over hidden states. These components are not independent modules that happen to be connected, but mutually constraining parts of a single generative architecture. The observation model determines what evidence would be expected under different hidden states, the transition model determines how those states are expected to change, prior preferences determine which outcomes are desirable and policy beliefs determine which actions are likely to realise informative or preferred futures.

For the purposes of this paper, the key implication is that active inference gives a principled way to build world models that are simultaneously predictive, epistemic and action-oriented. Such models do represent what hidden states might be causing current observations, what actions would reveal uncertain aspects of the world and what futures should be actively brought about.

3.1 Active inference world models

We define an active inference world model as a generative model over hidden states, observations, actions, policies and preferred outcomes. In its simplest form, the agent assumes that observations o_t are generated by hidden states s_t , that hidden states evolve under policies π and that outcomes can be evaluated relative to prior preferences C . A compact formulation is:

$$P(o_{1:T}, s_{1:T}, \pi) = P(\pi)P(s_1) \prod_{t=1}^T P(o_t | s_t)P(s_{t+1} | s_t, \pi). \quad (1)$$

Here, $P(o_t | s_t)$ is the observation model, $P(s_{t+1} | s_t, \pi)$ is the transition model and $P(\pi)$ encodes prior beliefs over policies. In active inference, preferences are usually specified as prior beliefs over

outcomes, such that preferred observations are assigned higher prior probability:

$$P(o_t | C) \propto \exp(C(o_t)). \quad (2)$$

Perceptual inference then corresponds to estimating posterior beliefs $Q(s)$ by minimising variational free energy. This objective balances the accuracy of the agent’s explanation against the complexity of moving away from prior beliefs:

$$F = \underbrace{D_{\text{KL}} [Q(s) \| P(s)]}_{\text{complexity}} - \underbrace{\mathbb{E}_{Q(s)} [\ln P(o | s)]}_{\text{accuracy}}. \quad (3)$$

The accuracy term increases when the inferred hidden states make the observations likely, while the complexity term penalises posterior beliefs that depart too far from the prior. Minimising free energy therefore favours explanations that fit the data well, but only when the improvement in fit justifies the additional complexity.

Policy selection is governed by expected free energy, which evaluates policies according to the observations and beliefs they are expected to produce in the future:

$$G(\pi) = \mathbb{E}_{Q(o_\tau, s_\tau | \pi)} [\ln Q(s_\tau | \pi) - \ln P(o_\tau, s_\tau | C)]. \quad (4)$$

Although different decompositions are possible, a useful interpretation is that expected free energy combines pragmatic value with epistemic value. Policy beliefs can then be written as a softmax over expected free energy:

$$Q(\pi) = \sigma(-\gamma G(\pi)), \quad (5)$$

where γ is a policy precision parameter that controls how strongly differences in expected free energy influence action selection.

4 A toy-house agent

To make the active inference world model concrete, we implemented a minimal object-search agent in a simulated house environment. The aim of this example is to isolate the inferential structure of the problem: the agent must search for a hidden target object, use prior expectations about where that object is likely to occur and update its beliefs as observations become available. The implementation therefore abstracts away from detailed locomotor control and treats the agent as a floating ray-cast sensing platform which allows the example to focus on viewpoint selection, semantic priors and belief updating.

Environment and observations.

The environment consists of a simple house-like layout containing rooms, furniture, candidate observation sites and a hidden target object. At any point in time, the true object location is

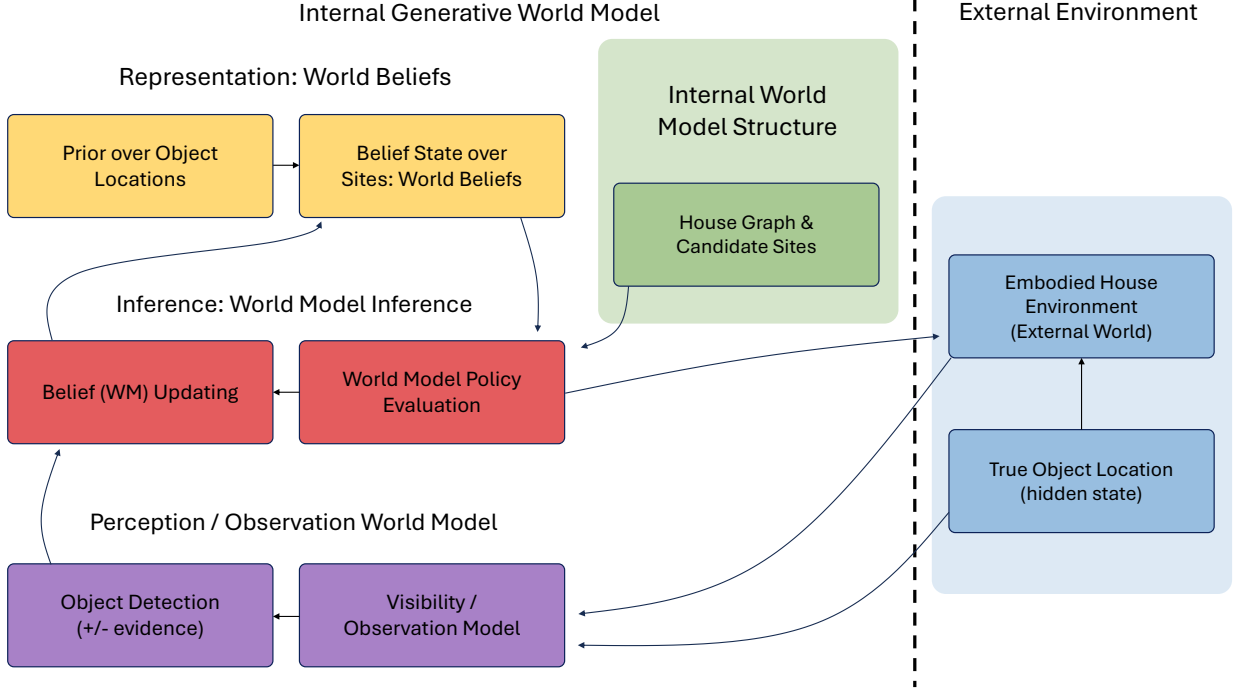


Figure 1: Schematic overview of the toy-house active inference world model. The agent maintains an internal generative world model comprising prior beliefs over object locations, belief states over candidate sites, an internal model of the house structure, policy evaluation, belief updating and an observation model. Policies are evaluated over candidate sites and routes through the house graph, then enacted in the external embodied house environment. The external environment contains the true object location as a hidden state. Sensory evidence, generated through visibility and object-detection processes, is passed through the observation model and used to update beliefs over possible object locations.

fixed but unobserved by the agent. The agent occupies a position in the environment and receives sensory evidence through ray-cast observations, which report whether the target is visible from the current viewpoint. This gives a sparse observation model: the agent does not receive a full semantic map or object-labelled image and only action-dependent evidence that can confirm or disconfirm candidate object locations.

Thus, the agent cannot infer the hidden state of the world from a single static input because what it observes depends on where it chooses to move and what can be seen from that location. A failed observation is therefore informative only when the agent expected that the object would have been visible if it were present at a particular candidate site. In this sense, the observation model must encode both positive evidence, when the object is detected and negative evidence, when a viewpoint rules out locations that should have been visible.

Beliefs and semantic priors.

The agent maintains a belief distribution over possible object locations. Before any observations are made, this distribution is not uniform because the agent is equipped with simple semantic priors about where different objects are likely to be found. These priors are minimal structured

expectations that link object identity to likely regions of the environment. For example, an object associated with work or study may be assigned higher prior probability near a desk, while an object associated with cooking may be assigned higher prior probability in a kitchen-like region.

These semantic priors play the role of structured prior beliefs over hidden states. They allow the agent to begin search from a plausible hypothesis space. Importantly, however, the priors do not determine behaviour on their own. They are revised as sensory evidence accumulates, so that a highly plausible location can be down-weighted if the agent samples an informative viewpoint and fails to observe the target. This is the key difference between a semantic heuristic and a generative belief system, meaning, the prior guides search but remains answerable to evidence.

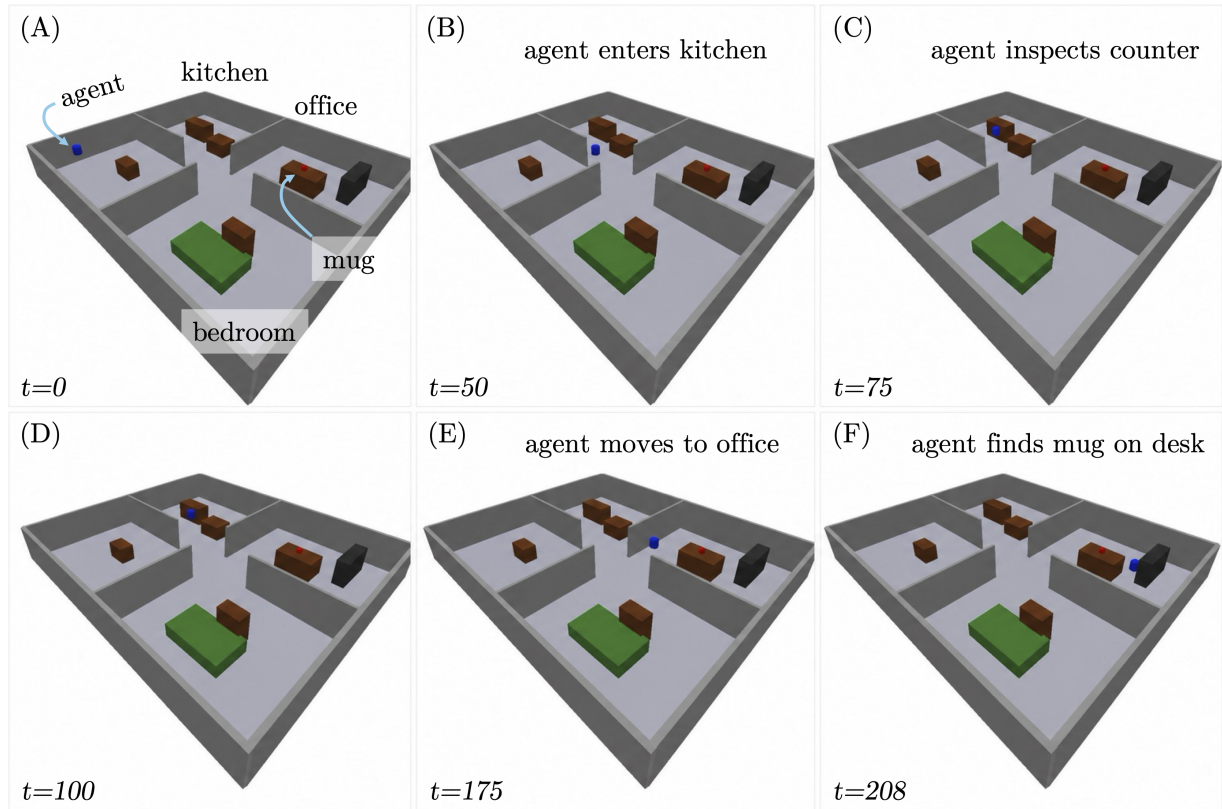


Figure 2: Semantic search and belief revision in the toy-house active inference world model. The agent is tasked with finding a mug in a simplified household environment containing a kitchen, office and bedroom. (A) At $t = 0$, the agent begins with a semantic prior that mugs are most likely to be found in the kitchen or office, with the kitchen initially assigned higher probability because mugs are commonly associated with counters and food-preparation spaces. (B) Guided by this prior, the agent first navigates to the kitchen. (C) It then inspects the kitchen counter, sampling the location with the highest expected information gain under the current belief state. (D) After failing to observe the mug in the kitchen, the agent updates its posterior beliefs, reducing the probability that the mug is in the kitchen and increasing the relative probability of alternative semantically plausible locations. (E) The revised belief state redirects behaviour toward the office. (F) The agent finds the mug on the office desk. The blue marker denotes the agent and the red marker denotes the mug. These frames are from the full PyBullet physics-world simulation of the active inference agent.

Policy selection and belief updating.

At each decision point, the agent evaluates candidate policies corresponding to movements toward possible observation sites. Each policy is scored according to a simple expected free energy objective, which combines the pragmatic value of finding the target with the epistemic value of reducing uncertainty over possible object locations. Policies that move the agent toward highly probable candidate locations are favoured because they may lead to target discovery, while policies that allow the agent to rule out several plausible locations are also favoured because they are informative even if the target is not immediately found.

After executing a policy, the agent receives a ray-cast observation and updates its beliefs over candidate object locations. If the object is detected, posterior probability concentrates on the observed location. If the object is not detected, the agent reduces the probability assigned to locations that should have been visible from the sampled viewpoint, while preserving probability mass over locations that remain unobserved or occluded. The agent therefore accumulates evidence through both confirmation and exclusion, which allows search behaviour to become progressively more focused as uncertainty is resolved.

5 Relationship to reinforcement learning world models

Reinforcement learning world models and active inference world models share an important starting point: both reject the idea that intelligent behaviour can be understood as a direct mapping from sensory input to action. In both cases, the agent benefits from an internal model that can be used to estimate the consequences of possible actions before those actions are taken. This is the central reason why world models have become so influential in model-based reinforcement learning. If an agent can learn a latent model of environmental dynamics, then it can use that model to plan, imagine rollouts, improve policies and reduce the amount of costly interaction required in the external environment [Ha and Schmidhuber, 2018, Hafner et al., 2023]. In this respect, reinforcement learning world models have provided some of the clearest computational demonstrations that internal simulation can support flexible adaptive behaviour.

The distinction between active inference and reinforcement learning is the role that the model plays in the architecture of the agent. In many reinforcement learning systems, the world model is learned as a predictive substrate for policy optimisation. The model estimates future latent states, observations, rewards or continuation signals, while a policy or value function is trained to select actions that maximise expected return under those predictions. This division is often highly effective, especially in domains where rewards are well defined and training can be organised around repeated interaction with an environment. However, it also means that the model, the reward function and the policy objective remain conceptually separable, even when they are trained jointly in practice.

Active inference changes the interpretation of this architecture by treating perception, action and policy selection as parts of one inferential process. The agent maintains posterior beliefs about hidden states and selects policies according to the expected free energy of future beliefs and observations. This means that the objective already contains both preference-seeking and uncertainty-reducing components. What reinforcement learning often handles through reward shaping, intrinsic motivation, exploration bonuses or auxiliary objectives is treated in active inference as part of the same generative model that explains sensory data and evaluates possible futures.

The difference becomes especially clear in object search, where a reinforcement learning formulation would typically define the task in relation to an external reward, such as a positive signal for finding the mug, perhaps combined with a cost for time, movement or failed search. A learned world model could then support behaviour by predicting which trajectories are likely to produce that reward, allowing the agent to evaluate possible futures before acting. In the active inference formulation, however, the same situation is cast in explicitly inferential terms: the mug location is

a hidden state, the agent’s observations are evidence about that hidden state and successful search corresponds to the progressive resolution of uncertainty under prior preferences. Moving to the kitchen is therefore not only a step toward a possible reward, but an epistemic intervention that tests a plausible hypothesis about where the mug might be, while failing to see the mug is not simply a non-rewarded transition, but negative evidence that reshapes the posterior distribution over possible locations. The same behavioural sequence can therefore be described within both frameworks, but active inference changes the interpretation of what the agent is doing, from seeking rewarded transitions to actively sampling the world in order to refine a generative model.

This distinction also affects how semantic structure enters the architecture. In a reinforcement learning agent, expectations about mugs, kitchens, desks and offices might be acquired implicitly through many training episodes, encoded in a latent representation, supplied through language conditioning or embedded indirectly in the reward and policy architecture. In an active inference world model, by contrast, the same structure can be expressed directly as a prior over hidden states, meaning that the agent begins with beliefs about where an object is likely to be, uses action to sample evidence from the environment and then revises those beliefs when observations contradict the prior. Semantic knowledge therefore has a precise computational role: it constrains the hypothesis space for inference, so that search begins from plausible possibilities, while still remaining answerable to sensory evidence rather than becoming a rigid heuristic.

The two approaches should therefore not be treated as mutually exclusive, since many modern reinforcement learning systems already contain ingredients that resonate strongly with active inference, including latent dynamics, uncertainty-aware prediction, imagination-based planning and intrinsic objectives for exploration. Conversely, active inference agents can be implemented using many of the same computational tools that have been developed in reinforcement learning, including learned latent representations, amortised inference, differentiable transition models and scalable policy search. The main contribution of active inference is therefore to reorganise these tools within a normative architecture in which generative modelling, belief updating, prior preferences and expected information gain are treated as interdependent components of adaptive behaviour.

From this perspective, active inference world models offer a complementary route for developing the next generation of model-based agents. Reinforcement learning has shown that learned world models can support powerful forms of planning and control, while active inference suggests how such models might be extended into embodied belief systems that explicitly represent hidden causes, uncertainty, epistemic value and preference.

6 Toward general-purpose adaptive intelligence

The toy-house agent is a minimal example, but it illustrates a more general architectural direction. If agents are to operate flexibly in open-ended environments, they will need models that do more than predict future observations. They will need models that help them decide what to sample, what to rule out and how to revise their assumptions when the world fails to match expectation.

From this perspective, general-purpose adaptive intelligence is a matter of coupling predictions more tightly through action. Active inference offers one way to do this because it treats world

modelling as an ongoing process of embodied belief revision, in which behaviour is guided by both what the agent wants to achieve and what it still needs to know.

7 Conclusion

Here we argue that active inference provides a useful framework for rethinking world models as embodied generative systems. The toy-house agent makes this idea concrete by showing how semantic priors, action-dependent observations and belief updating can support a simple but flexible form of object search.

The broader point is that world models become more powerful when they are not only used to predict possible futures, but to guide the sampling of evidence that changes what the agent believes. In this sense, active inference reframes adaptive intelligence as the ability to act through a model, learn through action and continually reshape behaviour as uncertainty is resolved.

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